

Artificial Intelligence

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Lecture 5

Logical Agents

(Propositional Logic)

- Knowledge-based agents
- Propositional logic
- Semantic of propositional logic
- Proof by inference
- Proof by resolution

Knowledge-based agents

- A knowledge-based agent utilizes its **knowledge base (KB)** for its **reasoning** processes.
- **Knowledge base (KB)** is composed of a set of **sentences**.
- **Sentence** represents the **fact** about the world.

Example of KB:

“ $X > Y$ ” represents “X is greater than Y”

“ $X \neq Y$ ” represents “X is not equal to Y”

- A knowledge-based agent operates by ‘**TELL**’ and ‘**ASK**’ methods.
- **TELL**: add a new sentence to KB.
- **ASK**: query what is known in KB.

Fact: 9 is greater than 6

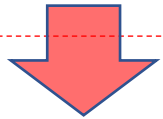
TELL

KB

Rule: If ($> x y$) and ($> y z$) then ($> x z$)

Sentence: ($> 9 6$)

Sentence: ($> 6 3$)



Reasoning (Inference)

New sentence: ($> 9 3$)

Fact: 6 is greater than 3

TELL

($> 9 ?$)

ASK

? = 3, 6

function KB-AGENT(*percept*) **returns** an *action*

persistent: *KB*, a knowledge base

t, a counter, initially 0, indicating time

Constructs a sentence asserting that the agent perceived the given percept at the given time.

Ex: Percept(Bump, *t*)

TELL(*KB*, MAKE-PERCEPT-SENTENCE(*percept*, *t*))

action ← ASK(*KB*, MAKE-ACTION-QUERY(*t*))

Constructs a sentence that asks what action should be done at the current time.

Ex: Action(?, *t*)

TELL(*KB*, MAKE-ACTION-SENTENCE(*action*, *t*))

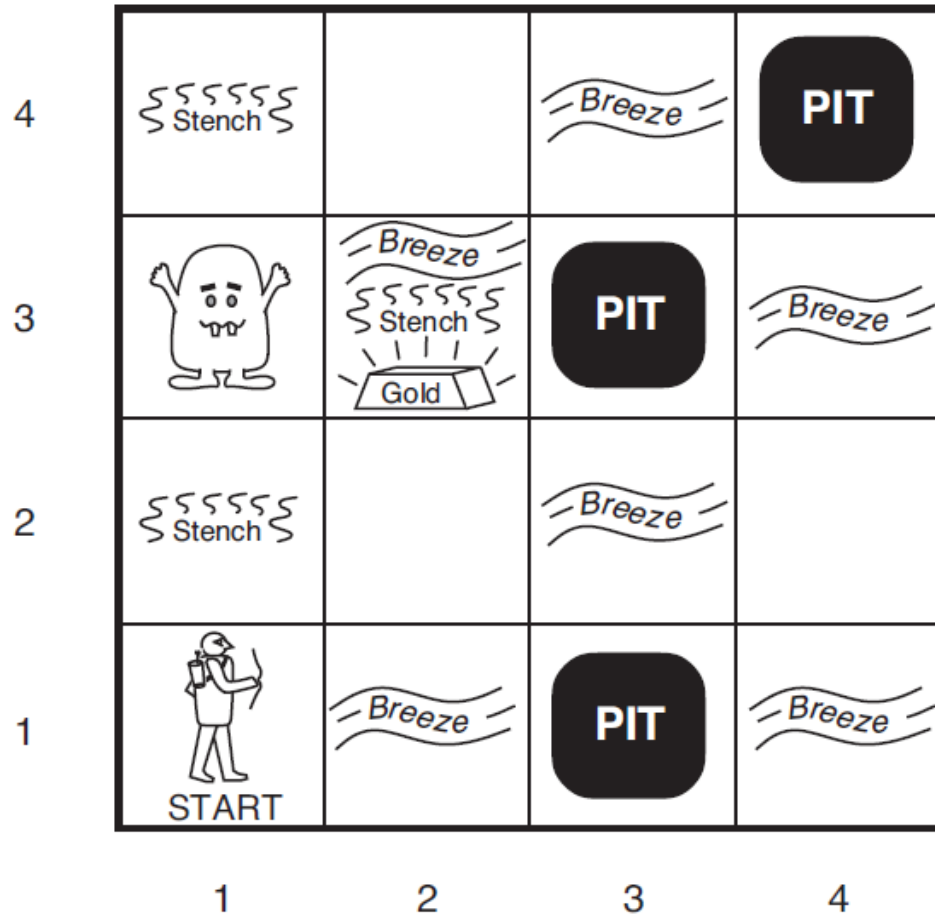
t ← *t* + 1

return *action*

Constructs a sentence asserting that the chosen action was executed.

Ex: AgentDid(TurnRight, *t*)

Case study : The Wumpus World



- The Wumpus world is a cave consisting of 4x4 rooms.
- **Wumpus** is a beast that cannot move but eats anyone entering its room.
- The Wumpus can be shot by an agent, but the agent has only one **arrow**.
- Some rooms contain bottomless **pits** that will trap anyone who wanders into these rooms.
- The agent must find a path leading to a **gold** and safely return to the origin (1,1).
- Actions: { **Forward, TurnLeft, TurnRight, Grab, Shoot, Climb** }
- Percepts: { **Stench, Breeze, Glitter, Bump, Scream** }

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2	2,2	3,2	4,2
1,1 A OK	2,1 OK	3,1	4,1

(a)

A = Agent
B = Breeze
G = Glitter, Gold
OK = Safe square
P = Pit
S = Stench
V = Visited
W = Wumpus

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2 OK	2,2 P?	3,2	4,2
1,1 V OK	2,1 A B OK	3,1 P?	4,1

(b)

The agent will turn around, go back to [1,1], and then proceed to [1,2]

Assume that the agent turns and moves to [2,3].

1,4	2,4	3,4	4,4
1,3 W!	2,3	3,3	4,3
1,2 A S OK	2,2 OK	3,2	4,2
1,1 V OK	2,1 B V OK	3,1 P!	4,1

(a)

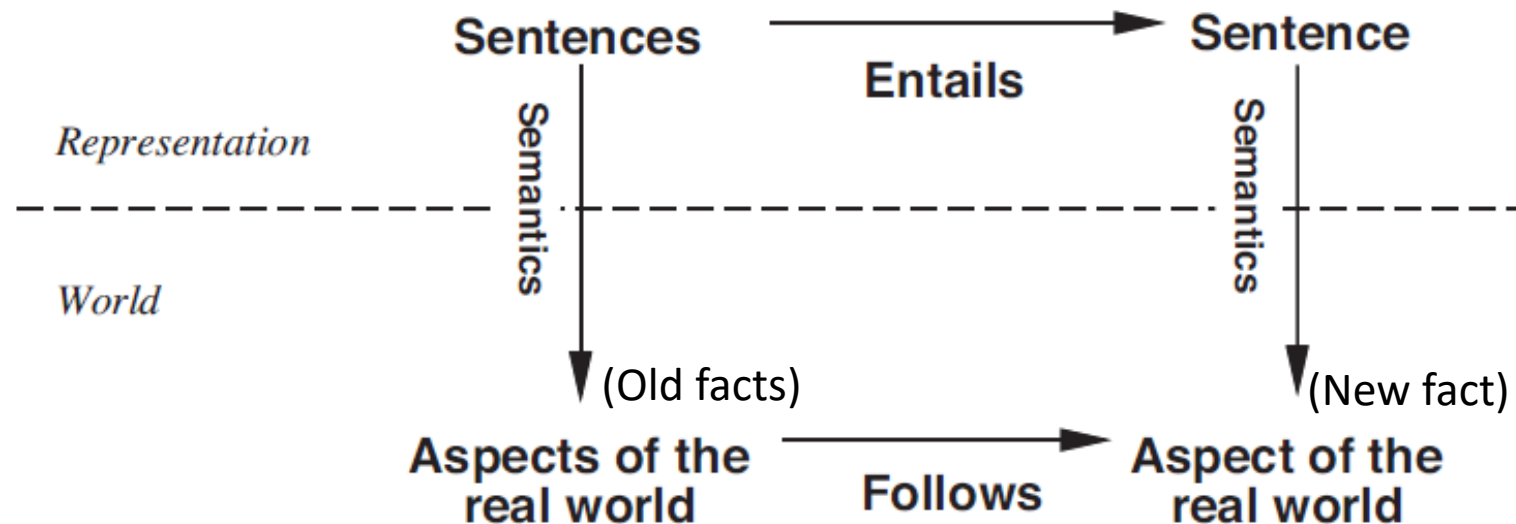
A = Agent
B = Breeze
G = Glitter, Gold
OK = Safe square
P = Pit
S = Stench
V = Visited
W = Wumpus

1,4	2,4 P?	3,4	4,4
1,3 W!	2,3 A S G B	3,3 P?	4,3
1,2 S V OK	2,2 V OK	3,2	4,2
1,1 V OK	2,1 B V OK	3,1 P!	4,1

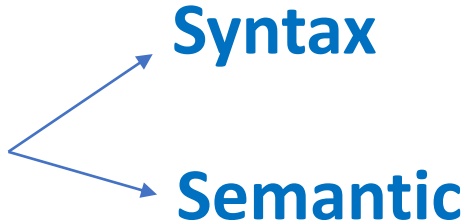
(b)

Reasoning

- **Reasoning** is the process of constructing new sentences from the old ones.
- A new sentence must **entail** the old sentences
- Hence, The fact of a new sentence must **follow** the old facts.



Propositional Logic: A Very Simple Logic



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graph LR; A[Propositional Logic: A Very Simple Logic] --> B[Syntax]; A --> C[Semantic];
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Syntax: (How a sentence is defined.)

- The **atomic sentences** consist of a single symbol.
 - **Symbol** represents the fact that can be true or false.
For example, $W_{1,3}$ stands for the proposition that the wumpus is in $[1,3]$.
- Symbol also includes two logical constants: **True** and **False**.

- **Complex sentences** are constructed from simpler sentences, using parentheses and logical connectives.

There are five logical connectives in common use:

- **$\neg$ (not)** A sentence such as $\neg W_{1,3}$ is called the **negation** of $W_{1,3}$.
- **$\wedge$ (and)** A sentence whose main connective is $\wedge$, such as $W_{1,3} \wedge P_{3,1}$, is called a **conjunction**.
- **$\vee$ (or)** A sentence using $\vee$, such as $(W_{1,3} \wedge P_{3,1}) \vee W_{2,2}$, is a **disjunction** of the disjuncts $(W_{1,3} \wedge P_{3,1})$ and $W_{2,2}$.
- **$\Rightarrow$ (implies)** A sentence such as $(W_{1,3} \wedge P_{3,1}) \Rightarrow \neg W_{2,2}$ is called an **implication** (or conditional). Its **premise** is $(W_{1,3} \wedge P_{3,1})$, and its **conclusion** is $\neg W_{2,2}$.
- **$\Leftrightarrow$ (if and only if)** The sentence $W_{1,3} \Leftrightarrow \neg W_{2,2}$ is a **biconditional**. Some other books write this as $\equiv$.

$$\begin{aligned}
\textit{Sentence} &\rightarrow \textit{AtomicSentence} \mid \textit{ComplexSentence} \\
\textit{AtomicSentence} &\rightarrow \textit{True} \mid \textit{False} \mid P \mid Q \mid R \mid \dots \\
\textit{ComplexSentence} &\rightarrow (\textit{Sentence}) \mid [\textit{Sentence}] \\
&\mid \neg \textit{Sentence} \\
&\mid \textit{Sentence} \wedge \textit{Sentence} \\
&\mid \textit{Sentence} \vee \textit{Sentence} \\
&\mid \textit{Sentence} \Rightarrow \textit{Sentence} \\
&\mid \textit{Sentence} \Leftrightarrow \textit{Sentence}
\end{aligned}$$

OPERATOR PRECEDENCE : $\neg, \wedge, \vee, \Rightarrow, \Leftrightarrow$

Figure 7.7 A BNF (Backus–Naur Form) grammar of sentences in propositional logic, along with operator precedences, from highest to lowest.

Semantics

- The semantics explain the truth of a sentence.
- A **truth table** can be used as the semantics tool.

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
<i>false</i>	<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>	<i>true</i>	<i>true</i>
<i>false</i>	<i>true</i>	<i>true</i>	<i>false</i>	<i>true</i>	<i>true</i>	<i>false</i>
<i>true</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>
<i>true</i>	<i>true</i>	<i>false</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>

- For example, the sentence $\neg P_{1,2} \wedge (P_{2,2} \vee P_{3,1})$
when $P_{1,2} = \text{false}$, $P_{2,2} = \text{false}$, $P_{3,1} = \text{true}$, gives
 $\text{true} \wedge (\text{false} \vee \text{true}) = \text{true} \wedge \text{true} = \text{true}$.

Seven inference rules for propositional logic

Premise
Conclusion

- Modus Ponens

$$\frac{\alpha \rightarrow \beta, \alpha}{\beta}$$

- And-Elimination

$$\frac{\alpha_1 \wedge \alpha_2 \wedge \dots \wedge \alpha_n}{\alpha_i} ; 1 \leq i \leq n$$

- And-Introduction

$$\frac{\alpha_1, \alpha_2, \dots, \alpha_n}{\alpha_1 \wedge \alpha_2 \wedge \dots \wedge \alpha_n}$$

- Or-Introduction

$$\frac{\alpha_i}{\alpha_1 \vee \alpha_2 \vee \dots \vee \alpha_n} ; 1 \leq i \leq n$$

- Double-negation Elimination

$$\frac{\neg \neg \alpha}{\alpha}$$

- Unit Resolution

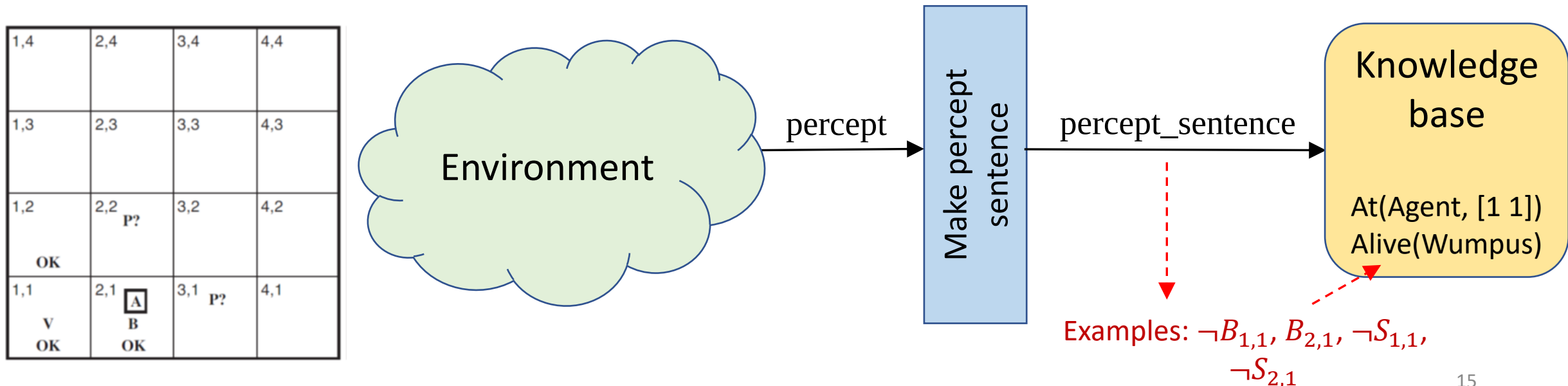
$$\frac{\alpha \vee \beta, \neg \beta}{\alpha}$$

- Resolution

$$\frac{\alpha \vee \beta, \neg \beta \vee \gamma}{\alpha \vee \gamma}$$

Inference in Wumpus World

- The agent must start with some knowledge of the environment.
For example, the agent's position is [1,1], and Wumpus is alive.
- Knowledge base is told new percept sentences or inferred sentences at each time step.



Example

- We will prove $\neg P_{1,2}$, i.e., there is **no pit in [1,2]**.
- We start with the knowledge base containing R1 through R5

$$R_1 : \quad \neg P_{1,1} .$$

$$R_2 : \quad B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1}) .$$

$$R_3 : \quad B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1}) .$$

$$R_4 : \quad \neg B_{1,1} .$$

$$R_5 : \quad B_{2,1} .$$

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2 OK	2,2 P?	3,2	4,2
1,1 V OK	2,1 A B OK	3,1 P?	4,1

KB

$$R_1 : \neg P_{1,1} .$$

$$R_2 : B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1}) .$$

$$R_3 : B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1}) .$$

$$R_4 : \neg B_{1,1} .$$

$$R_5 : B_{2,1} .$$

Proof by inference $\neg P_{1,2}$: There is no pit in [1,2].

- First, we apply **biconditional elimination** to R2 to obtain:

$$R_6 : (B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1})) \wedge ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1}) .$$

- Then we apply **And-Elimination** to R6 to obtain:

$$R_7 : ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1}) .$$

- Logical equivalence for contrapositives gives:

$$R_8 : (\neg B_{1,1} \Rightarrow \neg(P_{1,2} \vee P_{2,1})) . \text{ This is called "inference".}$$

- Now we can apply Modus Ponens with R4 and R8.

$$R_9 : \neg(P_{1,2} \vee P_{2,1}) .$$

$$\text{Modus Ponens: } \frac{\alpha \rightarrow \beta, \alpha}{\beta}$$

- Finally, we apply De Morgan's rule and And-Elimination.

$$R_{10} : \neg P_{1,2} \wedge \neg P_{2,1} . \quad = \quad \neg P_{1,2} \quad \neg P_{2,1}$$

Proved.

Proof by resolution : We will prove $P_{3,1}$: There is **a pit at [3,1]**

Let us consider the steps in which the agent returns from [2,1] to [1,1] and then goes to [1,2], where it perceives no breeze. We add the following sentences to the KB:

R_{11} : $\neg B_{1,2}$ (TELL to KB during runtime)

R_{12} : $B_{1,2} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{1,3})$. (Preprogrammed in KB)

Apply **Biconditional elimination** to R12 to obtain:

$$(B_{1,2} \rightarrow P_{1,1} \vee P_{2,2} \vee P_{1,3}) \wedge (P_{1,1} \vee P_{2,2} \vee P_{1,3} \rightarrow B_{1,2})$$

Apply **And-Elimination**:

$$P_{1,1} \vee P_{2,2} \vee P_{1,3} \rightarrow B_{1,2}$$

Contrapositive:

$$\neg B_{1,2} \rightarrow \neg P_{1,1} \wedge \neg P_{2,2} \wedge \neg P_{1,3}$$

And-Elimination:

$$\begin{aligned} R_{13} &: \neg P_{2,2} . \\ R_{14} &: \neg P_{1,3} . \end{aligned}$$

1,4	2,4	3,4	4,4
1,3 W!	2,3	3,3	4,3
1,2 A S OK	2,2 OK	3,2	4,2
1,1 V OK	2,1 B V OK	3,1 P! P!	4,1

Proof by resolution : We will prove $P_{3,1}$: There is **a pit at [3,1]**

Apply **Biconditional elimination** to R3, followed by **Modus Ponens** with R5, to obtain R15

$$R_3 : B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1}) .$$

$$R_5 : B_{2,1} .$$

$$R_{15} : P_{1,1} \vee P_{2,2} \vee P_{3,1} .$$

1,4	2,4	3,4	4,4
1,3 W!	2,3	3,3	4,3
1,2 A S OK	2,2 OK	3,2	4,2
1,1 V OK	2,1 B V OK	3,1 P! 	4,1

$$\frac{\alpha \vee \beta, \neg \beta}{\alpha}$$

Now using **the unit resolution**: the literal $\neg P_{2,2}$ in R13 resolves with the literal $P_{2,2}$ in R15 to give the resolvent.

$$R_{13} : \neg P_{2,2} .$$

$$R_{15} : P_{1,1} \vee P_{2,2} \vee P_{3,1} .$$

$$R_{16} : P_{1,1} \vee P_{3,1} .$$

1,4	2,4	3,4	4,4
1,3 W!	2,3	3,3	4,3
1,2 A S OK	2,2 OK	3,2	4,2
1,1 V OK	2,1 B V OK	3,1 P!	4,1

Similarly, the literal $\neg P_{1,1}$ in R1 resolves with the literal $P_{1,1}$ in R16 to give

$$R_1 : \neg P_{1,1} .$$

$$R_{17} : P_{3,1} .$$

Proved.

Proof by resolution : $W_{1,3}$: There is the **Wumpus** at [1,3]

$$R_1 : \neg S_{1,1} \Rightarrow \neg W_{1,1} \wedge \neg W_{1,2} \wedge \neg W_{2,1}$$

$$R_2 : \neg S_{2,1} \Rightarrow \neg W_{1,1} \wedge \neg W_{2,1} \wedge \neg W_{2,2} \wedge \neg W_{3,1}$$

$$R_3 : \neg S_{1,2} \Rightarrow \neg W_{1,1} \wedge \neg W_{1,2} \wedge \neg W_{2,2} \wedge \neg W_{1,3}$$

$$R_4 : S_{1,2} \Rightarrow W_{1,3} \vee W_{1,2} \vee W_{2,2} \vee W_{1,1}$$

Finding the Wumpus:

1. Agent starts out at [1,1] and gets the percept $\neg S_{1,1}$. By using R1 and **Modus Ponens**, we obtain

$$\neg W_{1,1} \wedge \neg W_{1,2} \wedge \neg W_{2,1}$$

1,4	2,4	3,4	4,4
1,3 W!	2,3	3,3	4,3
1,2 A S OK	2,2 OK	3,2	4,2
1,1 V OK	2,1 B V OK	3,1 P!	4,1

2. Applying **And-Elimination**:

$$\neg W_{1,1} \quad \neg W_{1,2} \quad \neg W_{2,1}$$

3. Agent moves to [2,1] and gets percept $\neg S_{2,1}$.

By using R2, **Modus Ponens**, **And-Elimination**, we obtain

$$\neg W_{2,2} \quad \neg W_{2,1} \quad \neg W_{3,1} \quad \neg W_{1,1}$$

4. Agent moves to [1,1] then [1,2] and get $S_{1,2}$.

By using R4 and **Modus Ponens**, we obtain

$$W_{1,3} \vee W_{1,2} \vee W_{2,2} \vee W_{1,1}$$

5. Applying the **unit resolution**, where α is $W_{1,3} \vee W_{1,2} \vee W_{2,2}$ and β is $W_{1,1}$.

$$W_{1,3} \vee W_{1,2} \vee W_{2,2}$$

6. Applying the **unit resolution** again , where α is $W_{1,3} \vee W_{1,2}$ and β is $W_{2,2}$.

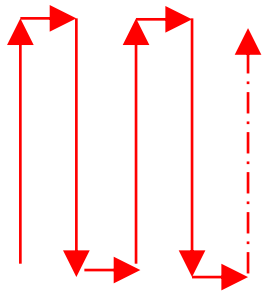
$$W_{1,3} \vee W_{1,2}$$

7. One more **unit resolution** where α is $W_{1,3}$ and β is $W_{1,3}$.

$$W_{1,3} \quad \boxed{\text{Proved.}}$$

1,4	2,4	3,4	4,4
1,3 W!	2,3	3,3	4,3
1,2 A S OK	2,2 OK	3,2	4,2
1,1 V OK	2,1 B V OK	3,1 P!	4,1

Exercise : Find a set of positions where agent visits.



Translating knowledge into action

We need additional rules that relate the current state of the world to the actions the agent should take.

For example,

$$A_{1,1} \wedge East_A \wedge \neg W_{2,1} \Rightarrow Forward$$

$$A_{1,2} \wedge North_A \wedge W_{1,3} \Rightarrow TurnRight$$

$$A_{1,2} \wedge North_A \wedge W_{1,3} \wedge Alive(W) \Rightarrow Shoot$$

Problems of the propositional agent

1. There are too many propositions to handle.

For example, The simple rule “Don’t go forward if the Wumpus is in front of the agent” can be stated in propositional logic by 64 rules.

2. Since the world can change configuration at each time step, we must specify **time** in the inference rules.

(Suppose that, when agent wants to turn, it always turns left.)

$$A_{1,1}^0 \wedge East_A^0 \wedge \neg W_{2,1} \Rightarrow Forward^0$$

$$A_{1,1}^6 \wedge East_A^6 \wedge \neg W_{2,1} \Rightarrow TurnLeft^6$$

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2	2,2	3,2	4,2
OK			
1,1	2,1	3,1	4,1
 OK	 OK		